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Toward Low-Cost yet Effective Temporal Learning for UAV Tracking

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Abstract

The utilization of temporal information has always been an open topic in the tracking community. However, existing trackers tend to employ more and more inputs or parameters for temporal learning, hindering their deployment in resource-constrained unmanned aerial vehicles (UAVs). More importantly, this raises ambiguity whether the performance gains come from the temporal learning itself, or come from the increased inputs and parameters. In this study, we advocate designing temporal learning components from a more balanced perspective that jointly considers performance gains and computational costs. To achieve this goal, we introduce a new evaluation metric, i.e., precision per FLOP (PPF). The PPF is introduced to quantify the tracking precision gains achieved by temporal learning components per unit of FLOP, thus enabling fair and efficiency-aware comparisons among these components and driving them toward more efficient designs. Based on this metric, we propose a low-cost yet effective temporal learning (LETL) approach to model contextual relationships. This approach continuously propagates and merges representative appearance tokens in video streams, allowing the tracker to efficiently capture the changing patterns of targets with relatively low costs. We integrate the LETL approach into existing one-stream frameworks, thereby building a simple yet effective tracker, namely LETrack, for robust UAV tracking. Extensive experimental demonstrate the superiority of our LETrack, and show that the proposed LETL approach achieves higher PPF scores. The Code is available at https://github.com/GXNU-ZhongLab/LETrack.

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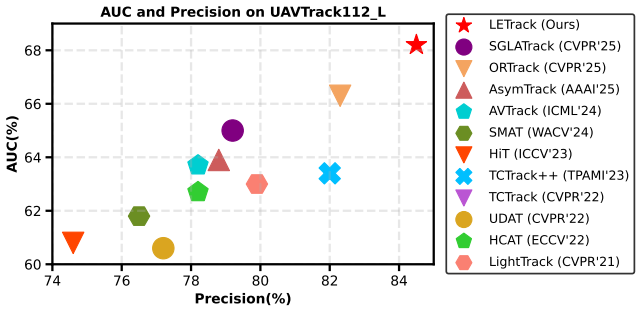


Figure 1. Comparison of AUC and precision between LETrack and other UAV trackers on UAVTrack112_L [18]. Our LETrack achieves SOTA performance in both AUC and precision.

1. Introduction

As an important branch of visual object tracking (VOT), unmanned aerial vehicle (UAV) tracking is widely used for various applications [37, 57]. Although UAV tracking has been investigated over the years, it remains an open task due to challenges such as camera motion, background clutter, similar interference, and other factors.

In recent years, the emergence of works such as AbaViTrack [33], AVTrack [35] and SGLATrack [56] has shifted the focus of UAV tracking research toward computational efficiency, thereby weakening the attention paid to the above-mentioned challenges to some extent. In fact, leveraging temporal information is an effective way to address these challenges, and it has been widely explored in the field of generic VOT. For example, STARK [58] and MixFormer [14] introduce an additional dynamic template beyond the initial one to capture appearance changes, while ODTrack [67] and MCITrack [27] further expand the number of templates to three and five, respectively. To better

integrate temporal information, EVPTrack [45] and AQATrack [54] replace the backbone with HiViT [65], whereas TemTrack [55] adopt Fast-iTPN [46] instead. Although these trackers achieve state-of-the-art (SOTA) performance, they often rely on multiple templates or larger backbones for temporal modeling, which significantly increases floating point operations (FLOPs) and hinders practical deployment for UAV tracking. More importantly, an interesting question arises: do their performance gains come from the temporal learning itself, or rather from increased FLOPs or larger backbones? This issue reveals a key gap in existing research. That is, while temporal learning is widely believed to improve tracking performance, the community still lacks a clear criterion to to decouple and evaluate the actual effectiveness of the temporal module.

To bridge this gap, we make a preliminary effort to explore the actual performance of temporal components under computational constraints, aiming to facilitate the development of efficient temporal learning strategies for UAV tracking. Considering the limited computational power of UAVs, an ideal temporal learning strategy for UAV tracking should achieve significant performance improvement at a low computational cost. On this basis, we introduce a new evaluation metric, precision per FLOP (PPF), specifically designed to quantify how much tracking precision a temporal learning strategy can improve per unit of FLOP. Using this metric, we evaluate the efficiency of temporal learning components in several well-known temporal trackers. Our analysis reveals that existing temporal learning strategies achieve relatively low PPF scores and are suboptimal for UAV tracking. To address this issue, we propose a low-cost yet efficient temporal learning (LETL) approach that can deliver significant performance improvement at negligible computational costs. The LETL approach extracts a group of informative appearance tokens as temporal tokens in each frame, and propagates them through the video stream to establish cross-frame contextual relationships. During propagation, similar temporal tokens from consecutive frames are merged to maintain efficiency. Benefiting from these designs, the LETL approach achieves higher PPF scores compared to other temporal learning strategies. We incorporate LETL into widely used one-stream frameworks, thereby building a simple yet effective tracker, i.e., LETrack, for robust UAV tracking. Extensive experiments demonstrate that our LETrack achieves state-of-the-art performance. Particularly, as shown in Fig. 1, LETrack attains 68.2% AUC and 84.8% precision on UAVTrack112_L [18], outperforming most existing UAV trackers. In summary, our main contributions are as follows:

- We propose to design temporal learning components from a balanced perspective that jointly considers performance gains and computational cost. On this basis, we explore a new evaluation metric to enable fair and efficiency-aware comparisons among these components, thereby driving temporal learning toward more efficient designs.
- We develop a low-cost yet effective temporal learning (LETL) approach according to the above perspective. This approach efficiently captures the changing patterns of object appearances in video streams with only negligible computational costs, providing novel insights.
- We integrate the LETL approach into existing one-stream frameworks, thereby building a simple yet effective tracker, namely LETrack, for robust UAV tracking. LETrack achieves state-of-the-art performance on six popular aerial datasets.

2. Related Works

2.1. UAV Tracking

As an essential subfield of VOT, UAV tracking demands trackers that can operate with low computational complexity on edge devices while remaining robust against the unique challenges posed by aerial platforms. Due to the efficiency, correlation filter (CF)-based trackers [1, 3, 15, 16, 23] have been widely used in UAV tracking. However, their robustness is often compromised under challenging conditions because of the limited representational capacity of hand-crafted features. Siamese network (SN)-based trackers [2, 29, 72] alleviate this issue via end-to-end deep learning, yet most of them still depend on early convolutional neural networks for feature extraction, leaving considerable room for further precision improvement. Recently, Vision transformers (ViTs) have been widely adopted across various domains due to their strong modeling capacity [12, 13, 28, 43, 62–64, 69, 70], and ViT-based trackers have also garnered increasing attention. To deploy ViTs on resource-constrained UAVs, many works have explored various approaches. For example, Aba-ViTrack [33] introduces an adaptive and background-aware token stopping method for tiny ViTs that learns to prune uninformative background tokens. Also based on tiny ViTs, AVTrack [35] employs activation modules to dynamically enable or disable ViT layers, further improving efficiency by avoiding extra unstructured access operations. SGLATrack [56] proposes a similarity-guided layer adaptation approach to achieve the trade-off between tracking precision and speed. ORTrack [52] designs an adaptive feature-based knowledge distillation approach to compress models. Although the above trackers have significantly promoted the progress of UAV tracking and demonstrated the potential of tiny ViTs in UAV tracking tasks, most of them place emphasis on improving computational efficiency. As a result, attention to the unique challenges posed by aerial platforms has been somewhat diminished. In this study, we propose LETrack to reawaken the attention to the robustness of UAV trackers. LETrack also builds upon tiny ViTs to maintain real-time

speed, but achieves better robustness via a low-cost yet effective temporal learning approach.

2.2. Temporal Learning in Tracking

Leveraging temporal information to improve tracking performance has always been a research focus in the community, regardless of whether the tracker is CF-based, SN-based, or ViT-based. Generally, there are two main strategies for exploiting temporal information. (i) Multi-template-based strategies. This type of strategy has a long history in the tracking community, remaining effective from traditional CF-based trackers to recent SN- and ViT-based ones. For example, CF-based trackers employ some confidence measures, like the peak-to-sidelobe ratio [22] and the average peak-to-correlation energy [47], to adjust the filter updating rate to adapt to appearance variations. Similarly, SN-based or ViT-based trackers, such as DROL[68], STARK [58], MixFormer [14], MCITrack [27], utilizes some scoring modules to maintain additional templates to improve the discriminability of the tracker. (ii) Trajectory-token-based strategies. This type of strategy has gained popularity with the emergence of ViTs. For example, ODTrack [67] and AQATrack [54] introduce learnable tokens or queries to compress intra-frame information and propagate them to establish inter-frame associations. Likewise, TemTrack [55] and MambaLCT [34] employ Mamba-based modules to model contextual relationships through the trajectory tokens. Although the above two strategies have achieved remarkable success in generic VOT, they may not be optimal for UAV tracking. Specifically, multi-template-based strategies can improve tracking performance to some extent, but they introduce considerable computational overhead. In contrast, trajectory-token-based strategies are relatively lightweight, but their performance improvements are limited, as the large proportion of background information resulting from UAV views dilutes the target features embedded in the trajectory tokens. Ideally, an excellent temporal learning strategy for UAV tracking should effectively enhance tracking performance while maintaining low computational cost. To this end, we propose PPF metric to enable efficiency-aware comparisons among these strategies and drive them toward more efficient designs. Note that while PPF has been explored in other fields [32], this is the first time it has been used in VOT to disentangle the impact of the temporal module.

3. Method

Here, we first give definition for pure trackers and temporal trackers in Sec. 3.1, thereby presenting the formulation of the PPF metric. Then, we provide a detailed description of the proposed low-cost yet effective temporal learning approach in Sec. 3.2. Finally, we summarize the overall pipeline of our LETrack in Sec. 3.3.

3.1. Evaluation for Temporal Components

Existing trackers typically incorporate temporal learning components to improve tracking performance, but only the overall tracking performance is compared in experimental evaluations. Such evaluations make it difficult to assess the actual performance of their introduced temporal learning strategies, thereby failing to provide clear guidance for designing efficient temporal components in resource-constrained UAV scenarios. In this subsection, we propose a new evaluation metric, i.e., precision per FLOP (PPF), specially designed to quantify the actual performance of temporal learning components within trackers. Before detailing the proposed metric, we first offer a review of deep learning (DL)-based trackers. Fundamentally, VOT can be regarded as a similarity matching problem between the template frame and the search frame. Therefore, a pure DL-based tracker takes only an initial template frame $\boldsymbol{z}$ and a current search frame $\boldsymbol{s}$ as input, formulated as:

$$\Psi(\boldsymbol{z}, \boldsymbol{s}) = \{\mathcal{G}(\boldsymbol{z}, \boldsymbol{s}), \mathcal{P}(\boldsymbol{s})\}, \tag{1}$$

where $\Psi(\boldsymbol{z}, \boldsymbol{s})$ denotes the pure tracker, $\mathcal{G}(\boldsymbol{z}, \boldsymbol{s})$ represents the backbone network responsible for feature extraction and interaction between $\boldsymbol{z}$ and $\boldsymbol{s}$, and $\mathcal{P}(\boldsymbol{s})$ is the prediction head for localizing the target in $\boldsymbol{s}$. To improve the performance of the pure tracker, a temporal learning component $\mathcal{T}(\boldsymbol{h})$ is introduced to capture the changing patterns of the target. Hence, the temporal version of $\Psi(\boldsymbol{z}, \boldsymbol{s})$ is denoted as:

$$\Psi(\boldsymbol{z}, \boldsymbol{s}, \boldsymbol{h}) = \{\mathcal{G}(\boldsymbol{z}, \boldsymbol{s}), \mathcal{P}(\boldsymbol{s}), \mathcal{T}(\boldsymbol{h})\}, \tag{2}$$

where $\boldsymbol{h}$ represents the historical information before the current frame. Let $\Psi_{\boldsymbol{z},\boldsymbol{s}}$ denote $\Psi(\boldsymbol{z}, \boldsymbol{s})$ and $\Psi_{\boldsymbol{z},\boldsymbol{s},\boldsymbol{h}}$ denote $\Psi(\boldsymbol{z}, \boldsymbol{s}, \boldsymbol{h})$ for simplicity, the PPF is defined as:

$$\text{PPF} = \frac{\Delta_{prec}}{\Delta_{flop}} = \frac{prec(\Psi_{\boldsymbol{z},\boldsymbol{s},\boldsymbol{h}}) - prec(\Psi_{\boldsymbol{z},\boldsymbol{s}})}{flop(\Psi_{\boldsymbol{z},\boldsymbol{s},\boldsymbol{h}}) - flop(\Psi_{\boldsymbol{z},\boldsymbol{s}})} \tag{3}$$

where $prec(\cdot)$ and $flop(\cdot)$ represent the functions used to compute the tracking precision and FLOPs of the tracker, respectively. For $\Psi_{\boldsymbol{z},\boldsymbol{s},\boldsymbol{h}}$ and $\Psi_{\boldsymbol{z},\boldsymbol{s}}$ in Eq. (3), the backbone and head networks remain unchanged, except for the temporal learning component. Consequently, the PPF metric enables a fair comparison of temporal learning components in trackers by isolating the influence of other factors. In addition, since some trackers have high computational complexity, the original form of PPF can become extremely small and meaningless. To address this, for the comparison where trackers have FLOPs $>$ 10G, the logarithmic form of PPF is used, i.e., PPF $= \Delta_{prec}/log(\Delta_{flop})$.

With the PPF metric, we conduct a study on several temporal-learning-based trackers, including ODTrack [67], HIPTrack [4], and EVPTrack [45]. Fig. 2 shows the overall precision $Prec$, improved precision Δ_{prec} and PPF scores

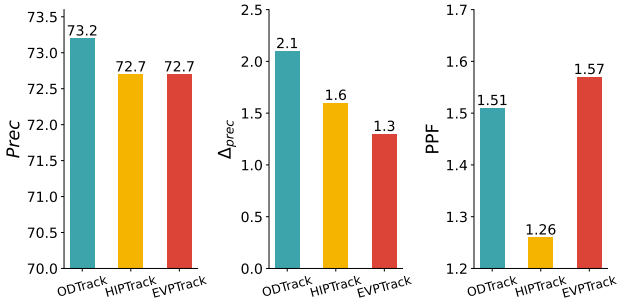


Figure 2. Comparison of overall precision $Prec$, improved precision Δ_{prec} and PPF for different temporal trackers on LaSOT [17].

of them. As can be seen, EVPTrack surpasses HIPTrack in $Prec$, yet Δ_{prec} brought by its temporal learning module is smaller. This is mainly because its superior performance benefits more from the stronger backbone (HiViT [65]) rather than from temporal learning itself. Moreover, although ODTrack achieves the largest Δ_{prec} through its temporal module, its PPF is lower than that of EVPTrack, as the use of three templates introduces substantial computational overhead. In fact, these trackers exhibit relatively low PPF scores, indicating that efficient temporal learning strategies are still far from being fully explored.

3.2. Low-Cost yet Effective Temporal Learning

Motivated by the above observations, we propose a low-cost yet effective temporal learning (LETL) approach that achieves notable performance gains without incurring excessive computational costs, making it more suitable for UAV tracking. The LETL is composed of a token extractor and a token combiner. The token extractor first collects a set of representative object appearance tokens (denoted as $\boldsymbol{h}^t$) from the current frame. The token combiner then mergers the similar tokens between $\boldsymbol{h}^t$ and previous-frame $\boldsymbol{h}^{t-1}$ to maintain efficiency. In the following, we provide a detailed description of the token extractor and the token combiner.

Token Extractor. This module collects several informative appearance tokens from the current frame to provide reliable references for target localization in subsequent frames. During the collection process, it considers both the similarity scores between the template and the search region and the classification scores from the prediction head. Specifically, let $\mathbf{Q}_z$, $\mathbf{K}_z$, $\mathbf{V}_z$, $\mathbf{Q}_s$, $\mathbf{K}_s$, $\mathbf{V}_s$ represent the queries, keys, and values of the template $\boldsymbol{z}$ and search $\boldsymbol{s}$, the similarity scores can be obtained from the standard one-stream attention mechanism, which is written as:

$$\begin{aligned} &Attn\left([\mathbf{Q}_z;\mathbf{Q}_s],[\mathbf{K}_z;\mathbf{K}_s],[\mathbf{V}_z;\mathbf{V}_s]\right) \\ &= \text{Softmax}\left(\begin{bmatrix}\mathbf{Q}_z\mathbf{K}_z^T & \mathbf{Q}_z\mathbf{K}_s^T \\ \mathbf{Q}_s\mathbf{K}_z^T & \mathbf{Q}_s\mathbf{K}_s^T\end{bmatrix}\right)\cdot\begin{bmatrix}\mathbf{V}_z \\ \mathbf{V}_s\end{bmatrix} \\ &= \begin{bmatrix}\mathbf{A}_{z2z} & \mathbf{A}_{z2s} \\ \mathbf{A}_{s2z} & \mathbf{A}_{s2s}\end{bmatrix}\cdot\begin{bmatrix}\mathbf{V}_z \\ \mathbf{V}_s\end{bmatrix}, \end{aligned} \tag{4}$$

where $[;]$ denotes the concatenation operation, and $\mathbf{A}_{s2z} \in \mathbb{R}^{N_s\times N_z}$ is the attention matrix representing the similarity between the search tokens and template tokens. We average matrix $\mathbf{A}_{s2z}$ across rows to compute the similarity score of each search token against the entire template, i.e., $\bar{\mathbf{A}}_{s2z} \in \mathbb{R}^{N_s}$, where N_s denotes the length of search tokens. To further improve the quality of the collected tokens, the classification scores from the prediction head are also considered, as they reflect the spatial distribution of the target. Hence, we flatten the 2D classification map, resulting in $\mathbf{C} \in \mathbb{R}^{N_s}$. With the similarity score $\bar{\mathbf{A}}_{s2z}$ and the classification score $\mathbf{C}$, the importance of each search token can be computed as:

$$\mathbf{S} = \frac{1}{M}\sum_{m=1}^{M}\{\bar{\mathbf{A}}_{s2z}\}^m \odot \mathbf{C} \tag{5}$$

where $\odot$ denotes the element-wise product, M is the number of attention heads, and $\mathbf{S}$ is the overall score. Based on $\mathbf{S}$, the top-K tokens with the highest scores are collected to construct representative token set $\boldsymbol{h}^t \in \mathbb{R}^{K\times D}$, where D is the embedding dimension.

Token Combiner. After extracting $\boldsymbol{h}^t$, the token combiner is enabled to merge it with previous-frame $\boldsymbol{h}^{t-1}$ for efficiency. First, it computes the pair-wise cosine similarity between the two sets $\boldsymbol{h}^t$ and $\boldsymbol{h}^{t-1}$. Then, each token in $\boldsymbol{h}^t$ is matched to its most similar counterpart in $\boldsymbol{h}^{t-1}$, and the top-r most similar matches are then selected, and the corresponding token pairs are merged. Since targets in video streams experience highly complex and diverse environmental conditions, making the quality of tokens extracted at later timesteps difficult to control. Hence, we prioritize preserving early extraced tokens and only merge a small subset of newly extracted ones. That is, the merged tokens from $\boldsymbol{h}^t$ are concatenated with the unmerged tokens from $\boldsymbol{h}^{t-1}$. Note that we use the token from $\boldsymbol{h}^t$ as the merged token's value, rather than averaging the tokens involved in the merge, which yields better performance in practice. We define this operation as:

$$\boldsymbol{h}^{t+1} = Merge(\boldsymbol{h}^t, \boldsymbol{h}^{t-1}, P) \tag{6}$$

where $\boldsymbol{h}^{t+1} \in \mathbb{R}^{K\times D}$ is the final output, and $P = r/K$ denotes the merging ratio.

3.3. LETrack for UAV Tracking

Overall Architecture. The overall architecture is shown in Fig. 3. Given a template image $\boldsymbol{z} \in \mathbb{R}^{3\times H_z\times W_z}$ and a search image $\boldsymbol{s} \in \mathbb{R}^{3\times H_s\times W_s}$, they are first tokenized into a template sequence $\boldsymbol{Z} \in \mathbb{R}^{N_z\times D}$ and a search sequence $\boldsymbol{S} \in \mathbb{R}^{N_s\times D}$. Then, $\boldsymbol{Z}$, $\boldsymbol{S}$ and $\boldsymbol{h}^{t-1}$ are concatenated, inputting to a one-stream encoder for unified feature extraction and interaction. Next, the encoded search features are employed by a prediction head to locate the target. After that, token

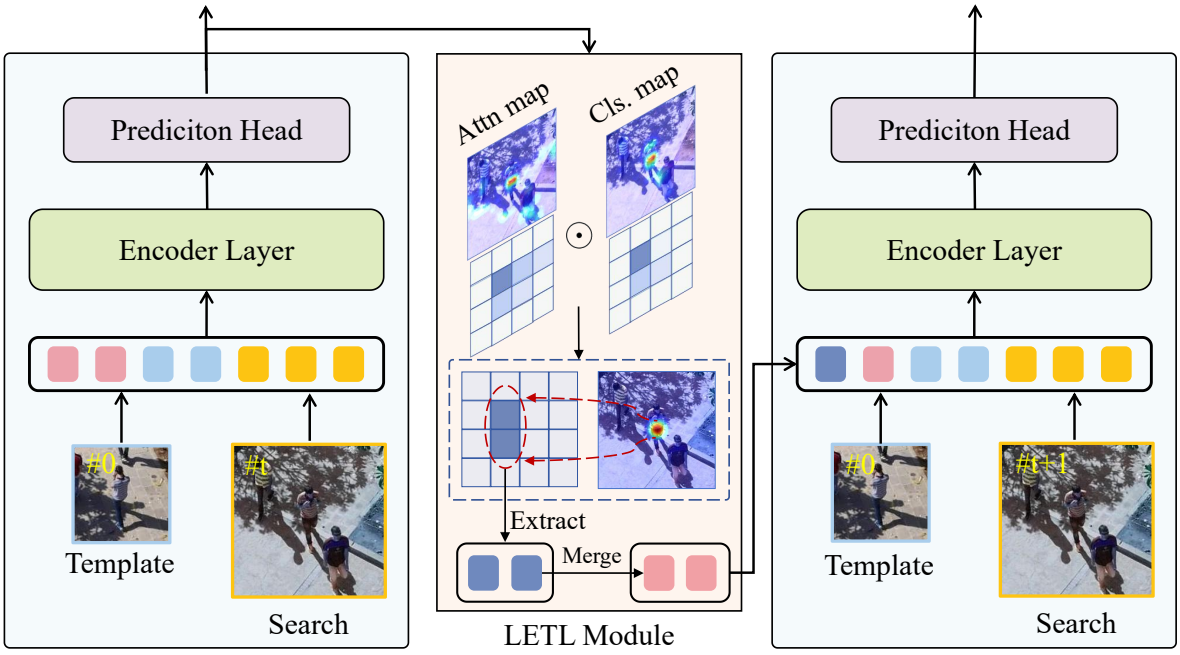


Figure 3. Overall architecture of the proposed LETrack. It consists of three parts, a one-stream encoder, a typical prediction head, and a low-cost yet effective temporal learning (LETL) module. The encoder leverages previous informative temporal tokens $\boldsymbol{h}^{t-1}$ to effectively extract features from the current search region. The prediction head takes these enhanced features to better localize the target. After completing the prediction for the current frame, the LETL module utilizes the attention maps from the encoder and the classification score maps from the prediction head, to extract newly informative tokens $\boldsymbol{h}^{t}$. It then merges $\boldsymbol{h}^{t}$ with similar elements in $\boldsymbol{h}^{t-1}$ to construct an updated temporal set $\boldsymbol{h}^{t+1}$, which is propagated to the next frame to guide subsequent tracking.

extractor is enabled to collect informative tokens $\boldsymbol{h}^{t}$ from the encoded search features. To maintain efficiency, token combiner is subsequently used to merge $\boldsymbol{h}^{t}$ and $\boldsymbol{h}^{t-1}$, i.e., $\boldsymbol{h}^{t+1} = Merge(\boldsymbol{h}^{t}, \boldsymbol{h}^{t-1}, P)$. Finally, $\boldsymbol{h}^{t+1}$ is propagated to the next frame for temporal modeling.

Head and Loss. Similar to most tracking methods [60, 67], we adopt a center-based prediction head to estimate the target's bounding box. The output tokens of the search features are firstly reshaped into a 2D spatial feature map to recover the spatial structure. These features are then fed into the prediction head to generate a local offsets, a bounding box size, and a classification score map. The target's position is determined by the location with the highest classification score, while the final tracking output is computed by integrating the offset and bounding box size. We adopt focal loss [39] for the classification task and a combination of L_1 loss and GIoU loss [44] for bounding box regression. The total loss $\mathcal{L}$ is calculated as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_{iou}\mathcal{L}_{iou} + \lambda_{L1}\mathcal{L}_{L1} \tag{7}$$

where $\lambda_{iou} = 2$ and $\lambda_{L1} = 5$ are the parameters.

4. Experiments

4.1. Implementation Details

Training Strategy. Similar to AVTrack [35] and SGLATrack [56], we employ DeiT-Tiny as the backbone encoder. We train LETrack on four widely used datasets: LaSOT [17], COCO [38], TrackingNet [42], and GOT-10k [24], with 1,000 videos excluded from GOT-10k as per the procedure in [58]. As in OSTrack [60], we apply common data augmentation techniques, and train LETrack with 300 epochs (60k matching pairs per epoch). We use the AdamW optimizer [40], setting the learning rate to 4×10^{-4} for the prediction head and 4×10^{-5} for the backbone. The learning rates are reduced by a factor of 10 after 240 epochs. All experiments are conducted on a server equipped with four NVIDIA Tesla V100 GPUs, with a batch size of 32.

Inference. LETrack takes a template image with 128×128 pixels and a search image with 256×256 pixels as the input. As the core of LETrack, the value for K and P in the LETL components are set to 16 and 37.5%, respectively. Moreover, We follows the common practice [8, 66] and uses the Hamming window to incorporate positional priors.

Table 1. AUC and precision comparisons with state-of-the-art trackers on UAVTrack112, UAVTrack112_L, UAV123, UAV123@10FPS and DTB70. The top two results are highlighted with red, blue, respectively.

Method	Source	UAVTrack112		UAVTrack112_L		UAV123		UAV123@10FPS		DTB70	
		AUC(%)	P(%)	AUC(%)	P(%)	AUC(%)	P(%)	AUC(%)	P(%)	AUC(%)	P(%)
LETrack	Ours	69.6	85.8	68.2	84.8	68.0	87.0	67.4	85.9	65.2	85.0
SGLATrack [56]	CVPR'25	67.5	82.8	64.0	79.2	66.9	84.9	65.5	82.6	65.1	84.4
ORTrack [52]	CVPR'25	66.6	82.1	65.3	82.3	66.1	84.0	64.1	80.1	65.1	83.7
AsymTrack-S [71]	AAAI'25	65.3	80.2	62.9	78.8	65.8	83.2	65.9	83.6	57.8	74.6
AVTrack [35]	ICML'24	65.4	80.3	62.7	78.2	66.8	84.8	65.8	83.2	65.0	84.3
SMAT [20]	WACV'24	65.3	82.2	60.8	76.5	64.6	81.8	63.5	80.4	63.8	81.9
HiT-Small [26]	ICCV'23	63.6	78.0	59.8	74.6	63.8	80.6	64.3	80.9	58.9	74.8
MVT [19]	BMVC'23	60.8	78.2	46.2	59.5	59.9	78.6	57.6	73.2	60.4	79.2
TCTrack++ [7]	TPAMI'23	64.0	81.5	62.4	82.0	59.3	78.0	59.9	78.0	62.6	81.5
TCTrack [6]	CVPR'22	59.4	76.6	58.3	78.6	60.5	80.0	58.7	77.4	62.2	81.2
UDAT [61]	CVPR'22	61.6	78.9	59.6	77.2	59.0	76.1	58.5	77.8	61.8	80.6
HCAT [9]	ECCV'22	66.0	81.9	61.7	78.2	64.8	84.2	64.4	81.9	63.7	83.5
LightTrack [59]	CVPR'21	63.8	81.1	62.0	79.9	62.7	78.3	59.9	75.1	59.1	76.4
HiFT [5]	ICCV'21	57.0	74.2	55.1	73.4	59.0	78.7	57.0	74.9	59.4	80.2
AutoTrack [36]	CVPR'20	46.5	69.4	40.2	67.5	47.2	68.9	47.7	67.1	47.8	71.6
ARCF [25]	ICCV'19	45.6	67.2	39.3	64.0	46.8	67.1	47.3	66.6	47.2	69.4
STRCF [30]	CVPR'18	43.0	63.5	36.0	60.9	48.1	68.1	45.7	62.7	43.7	64.9
MCCT-H [48]	CVPR'18	43.6	63.4	37.5	58.5	45.7	65.9	43.4	59.6	40.5	60.4
ECO-HC [15]	CVPR'17	47.4	68.6	41.7	64.8	49.6	71.0	46.8	64.0	44.8	63.5
fDSST [16]	TPAMI'17	39.1	56.8	30.6	49.1	40.5	58.3	37.9	51.6	35.7	53.4

Table 2. Comparison with state-of-the-art trackers on VisDrone2018. The top two results are highlighted with red, blue, respectively.

Method	SeqTrack CVPR'23[10]	SuperSBT TPAMI'24[53]	ACM-BAN TPAMI'24[21]	LiteTrack-B8 ICRA'24[49]	AVTrack ICML'24[35]	MCITrack-T224 AAAI'25[27]	SUTrack-T224 AAAI'25[11]	ORTrack CVPR'25[52]	SGLATrack CVPR'25[56]	LETrack Ours
AUC(%)	63.5	65.0	59.7	63.9	65.3	62.5	64.6	63.9	61.1	66.1
P(%)	82.3	84.9	79.4	83.2	86.0	79.4	82.8	84.6	79.8	87.2

4.2. Comparisons with the SOTA

UAVTrack112. UAVTrack112 [18] contains 112 challenging aerial tracking sequences that exhibit diverse challenges, such as low illumination, fast motion and low illumination scenarios. As shown in Tab. 1, our LETrack achieves the top-ranked performance on AUC and precision, reaching 69.6% and 85.8%, respectively. These evaluation results demonstrate that our LETrack, equipped with the LETL approach, effectively addresses the unique challenges posed by UAV platforms.

UAVTrack112_L. UAVTrack112_L [18] is currently the largest long-term aerial tracking benchmark, with more than 60K frames. Tab. 1 shows the results of the LETrack compared with other state-of-the-art trackers. Due to the proposed LETL approach, LETrack obtains the best performance. The outstanding performance on this benchmark proves the effectiveness of our tracker in extracting temporal cues for long-term tracking.

UAV123. UAV123 [41] is a large-scale and mainstream UAV tracking datasets consisting of 123 video sequences and over 112K frames. As shown in Tab. 1, our LETrack achieves a SOTA performance, surpassing other trackers.

UAV123@10FPS. UAV123@10FPS [41] is derived from the original UAV123 benchmark by down-sampling the frame rate from 30 FPS to 10 FPS. From Tab. 1, it can be observed that LETrack achieves SOTA performance with 67.4% AUC and 85.9% precision. This demonstrates that our LETL method can reliably model temporal information, significantly enhancing the tracker's robustness.

DTB70. DTB70 [31] includes 70 UAV-captured sequences, encompassing various challenges. As shown in Tab. 1, our LETrack also achieves the best results.

VisDrone2018. VisDrone2018 [51] is specifically designed to evaluate the robustness of trackers under UAV-specific challenges. As shown in Tab. 2, LETrack surpasses all the compared trackers, including some generic trackers such as SUTrack [11], MCITrack [27], and SeqTrack [10]. Notably, MCITrack, known for its temporal modeling capabilities, exhibits degraded performance on this challenging dataset. This is because its temporal learning strategy is suboptimal for UAV tracking scenarios, and is outperformed by our LETL approach.

4.3. Efficiency Evaluation

We evaluate the tracking efficiency of LETrack with several SOTA trackers, including SGLATrack [56], ORTrack [52], SeqTrack [10], and ARTrack [50] on a PC

Table 3. Comparison of FLOPs, model parameters, inference speed and tracking performance on UAVTrack112_L [18].

Trackers	AUC	FLOPs	Params	FPS
LETrack (Ours)	**68.2**	2.48	7.98	204
SGLATrack [56]	64.0	1.68	5.81	242
ORTrack [52]	65.3	1.54	5.31	288
SeqTrack [10]	67.1	65.86	89.11	43
ARTrack [50]	66.9	40.33	173.12	29

Table 4. Comparison of different temporal learning strategies.

Method	Δ_{prec}	Δ_{flop}	PPF
Baseline	(79.8)	(2.39)	-
+Dynamic template	+3.8	+0.40	9.5
+Dense video sampling	+5.9	+0.68	8.7
+Autoregressive queries	+4.3	+0.51	8.4
+LETL	+5.0	+0.09	55

machine equipped with an Intel Xeon 8480+ CPU and an NVIDIA A100 GPU. To ensure a fair comparison, all experiments are conducted under identical configurations, including the same Conda environment. As shown in Tab. 3, LETrack significantly outperforms mainstream trackers such as SeqTrack and ARTrack in both efficiency and precision, demonstrating its suitability for UAV tracking scenarios. Besides, since LETrack focuses on improving robustness rather than minimizing computational cost, its FLOPs are slightly higher than those of efficiency-oriented SGLATrack and ORTrack. Nevertheless, LETrack still runs at high speeds and can be further accelerated using the same inference optimization strategies as SGLATrack and ORTrack.

To further validate the tracking efficiency of our tracker, we deploy LETrack on the NVIDIA Jetson TX2 (4GB) embedded platform for real-world evaluation. Without using any acceleration techniques such as TensorRT, our tracker can run at a real-time speed of 26 FPS. In summary, These results demonstrate the efficiency of our tracker and its effectiveness in real-world applications.

4.4. Ablation Study

The core of our tracker lies in the LETL approach. To demonstrate its effectiveness, we conduct extensive ablation studies on UAVTrack112_L [18]. Note that default settings are marked in gray.

Comparisons with Other Strategies. To validate the superiority of our LETL approach, we compare it against other temporal learning strategies. Specifically, we first train a one-stream tracker using DeiT-Tiny as the backbone, which serves as the baseline. Then, We integrate various temporal modeling approaches into this baseline, including dynamic templates in Mixformer [14], dense template sequence sampling In ODTrack [67], autoregressive queries in AQATrack [54], and our proposed LETL. Tab. 4 illustrates the improved precision Δ_{prec}, the increased FLOPs Δ_{flop}, and the resulting PPF achieved by each temporal strategy over the baseline. As can be seen, the proposed LETL achieves the highest PPF, demonstrating its low-cost yet effective characteristic in resource-constrained UAV tracking scenarios. Although dense template sequence sampling yields the largest precision gain, it also introduces the highest computational overhead in terms of FLOPs, resulting in a lower PPF compared to our LETL approach. An interesting observation is that, although both autoregressive queries and LETL propagate target information sequentially across video frames, the former achieves a lower Δ_{prec} than LETL. We attribute this to the fact that autoregressive queries compress the entire search region into trajectory tokens. However, under UAV viewpoints, where background information often dominates the frame, which tends to dilute the target-specific information embedded in the trajectory tokens. In contrast, LETL propagates fine-grained local appearance tokens instead of compressed global representations, thereby delivering more performance gains.

Token Extractor and Token Combiner. The proposed LETL approach primarily consists of a token extractor and a token combiner. The core of the token extractor lies in capturing informative historical cues, while the token combiner focuses on merging redundant temporal tokens. Hence, we conduct a detailed investigation into both the token extraction and token merging methods, and the results are shown in Tab. 5. For the token extractor, the strategy that leverages both attention weights and classification scores to collect temporal tokens achieves the best performance. This validates the complementarity between attention weights and classification scores, i.e., the former captures similarity between the template and search frames, while the latter reflects the spatial likelihood of the target. For the token combiner, using the value of the more temporally recent token to represent the merged token yields better performance than simple averaging. This is because averaging tends to dilute fine-grained target features across frames and can even accumulate erroneous information over time, especially in cluttered UAV scenarios.

Extracted and Merged Tokens. There are two hyperparameters in the LETL approach, including extracted token number K and merging ratio P. In fact, K controls the size of the temporal token set propagated through the video, while P governs how frequently this set is updated in each frame. Tab. 6 illustrates how tracking performance varies with different choices for K and P. As can be seen, the best AUC and precision are achieved when K=32. Although K=16 yields slightly lower performance

Table 5. Ablation studies of the token extractor and combiner.

Module	Method	AUC	Prec.
Token Extractor	Attn	67.8	83.9
	Cls.	67.4	83.3
	Attn+Cls.	68.2	84.8
Token Combiner	Average	67.1	83.2
	Replace	68.2	84.8

Table 6. Ablation studies of the hyper-parameter K and P.

K	P	AUC	Prec.	PPF
32	50%	68.5	85.2	32
32	37.5%	68.8	85.7	35
16	50%	67.5	84.0	47
16	37.5%	68.2	84.8	55
16	25%	68.0	84.4	51
8	50%	65.2	81.2	33

than K=32, it attains the highest PPF, indicating the most favorable efficiency-precision trade-off. In contrast, performance drops significantly at K=8. Moreover, a merging ratio of P=37.5% yields the best performance. This is because an overly large ratio (50%) increases the risk of propagating erroneous information, while an overly small ratio (25%) fails to adequately capture appearance variations.

4.5. Qualitative Comparison and Visualization

To intuitively demonstrate the effectiveness of our tracker, we visualize the tracking results of LETrack and previous three SOTA models, including SGLATrack [56], ORTrack [52], and AVTrack [35]. As shown in Fig. 4, due to the excellent temporal modeling ability of our LETL approach, LETrack locates the target more accurately in these challenging sequences. In addition, we also visualize the attention maps of the tracker without and with the LETL module. As illustrated in Fig. 5, the tracker with the LETL module is more focused on the target under the distraction from similar objects. These results further validate the effectiveness of our LETL approach.

5. Conclusion

In this work, we identify a problem that has been largely overlooked by the community, i.e., the efficiency of temporal learning components within trackers. To this end, we introduce a new evaluation metric, PPF. PPF enables efficiency-aware comparisons among these components, thus driving them toward more computationally efficient designs. Guided by this metric, we propose a low-cost yet effective temporal learning approach. By continuously collecting, merging, and propagating informative appearance tokens across video frames, this approach significantly enhances the robustness of UAV trackers without compromising real-time speed. To validate its effectiveness, we integrate this approach into a pure one-stream framework, thus establishing LETrack. Experiments demonstrate that LETrack achieves SOTA performance.

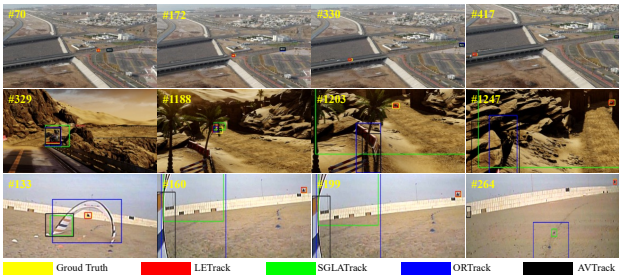


Figure 4. Qualitative comparisons of LETrack against other three SOTA trackers in UAV123 [41]. Best viewed by zooming in.

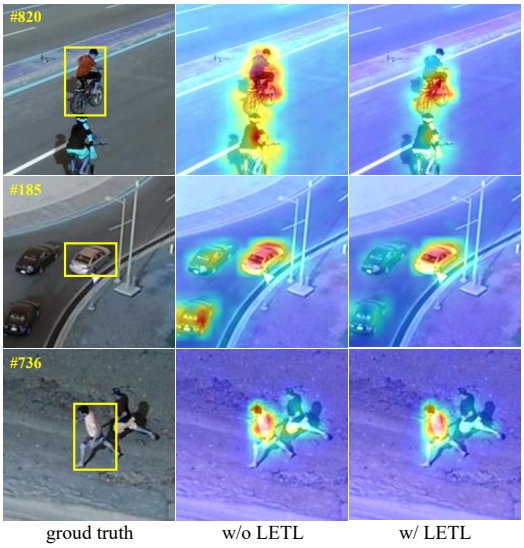


Figure 5. Comparison of attention maps. Note that w/o and w/ denote the tracker without and with LETL approach, respectively.

6. Acknowledgement

This work is supported by the Project of Guangxi Science and Technology (No. 2025GXNSFAA069676, 2024GXNSFGA010001, 2025GXNSFAA069417, and GuiKeFN2504240017), the National Natural Science Foundation of China (No.U23A20383, 62472109 and 62466051), the Guangxi "Young Bagui Scholar" Teams for Innovation and Research Project, the Research Project of Guangxi Normal University (No. 2025DF001).

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